

ARVIAT, NU, Canada

WMO: 711740

Lat: 61.098N

Lon: 94.069W

Elev: 34

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-38.9	-36.1	-47.0	0.4	-38.8	-44.1	0.4	-36.0	41.9	-17.4	36.8	-14.7	12.0	330	1.072

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	68.5	58.7	63.9	56.1	60.3	54.2	60.0	67.2	57.1	62.3	54.7	59.6	13.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.8	68.8	62.9	54.1	62.3	59.7	51.9	57.4	56.6	26.5	67.2	24.6	62.6	23.1	60.0	68.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.6	31.3	28.2	-41.2	78.6	4.4	4.2	-44.4	81.6	-46.9	84.1	-49.4	86.5	-52.5	89.6		
(5)			-41.4	65.5	4.3	2.1	-44.5	67.0	-47.0	68.2	-49.4	69.4	-52.5	70.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	-17.2	-18.3	-8.4	7.0	24.6	40.5	52.1	50.9	41.9	27.7	5.3	-9.2	(6)
(7)		DBStd	27.29	12.06	11.32	13.70	12.33	8.83	7.07	6.16	4.65	6.12	8.35	13.19	13.05	(7)
(8)		HDD50	12397	2082	1914	1811	1291	789	299	47	43	252	692	1341	1836	(8)
(9)		HDD65	17668	2547	2334	2276	1741	1254	736	402	436	693	1157	1791	2301	(9)
(10)		CDD50	205	0	0	0	0	0	13	111	72	9	0	0	0	(10)
(11)		CDD65	2	0	0	0	0	0	0	2	0	0	0	0	0	(11)
(12)		CDH74	33	0	0	0	0	0	7	24	2	0	0	0	0	(12)
(13)		CDH80	4	0	0	0	0	0	2	2	0	0	0	0	0	(13)
(14)	Wind	WSAvg	14.4	15.5	15.4	14.8	14.2	13.7	12.6	12.3	13.3	14.4	15.7	15.8	15.1	(14)
(15)	Precipitation	PrecAvg	12.6	0.6	0.5	1.0	0.7	0.7	1.1	1.6	2.2	1.9	1.2	0.8	0.6	(15)
(16)		PrecMax	19.8	1.9	3.0	6.1	2.4	1.9	2.2	2.9	4.1	5.0	3.7	3.8	2.0	(16)
(17)		PrecMin	6.6	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.4	0.4	0.3	0.0	0.1	(17)
(18)		PrecStd	3.6	0.6	0.6	1.4	0.6	0.5	0.5	0.7	1.0	1.0	0.8	0.8	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	15.3	28.8	35.1	47.8	72.1	77.9	71.9	64.2	47.7	33.3	25.8	(19)
(20)			MCWB	17.6	14.5	27.7	33.8	43.7	57.2	62.0	61.5	54.7	44.5	32.6	24.9	(20)
(21)		2%	DB	12.1	9.2	22.8	32.1	40.3	61.1	70.0	65.3	56.8	41.8	31.1	22.1	(21)
(22)			MCWB	11.5	8.6	22.2	31.3	37.8	53.3	59.5	57.4	51.3	40.2	30.4	21.5	(22)
(23)		5%	DB	6.4	5.1	17.6	28.8	37.0	54.6	66.1	61.6	53.0	39.4	29.0	16.3	(23)
(24)			MCWB	5.7	4.5	16.9	27.9	35.2	48.6	57.5	55.0	49.1	38.2	28.1	15.8	(24)
(25)		10%	DB	1.7	0.7	13.2	25.6	34.6	50.2	62.3	58.4	50.0	37.3	26.6	11.4	(25)
(26)			MCWB	1.0	0.0	12.6	24.7	33.4	45.4	55.5	53.3	47.5	36.1	25.7	10.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.7	15.1	28.1	34.2	43.5	58.9	64.3	63.2	57.1	45.1	32.9	25.4	(27)
(28)			MCDWB	17.8	15.7	28.9	35.0	47.6	70.0	72.5	68.6	62.8	47.5	33.4	25.9	(28)
(29)		2%	WB	11.6	8.7	22.3	31.4	37.9	53.5	61.1	58.7	52.1	40.6	30.5	21.5	(29)
(30)			MCDWB	12.0	9.1	22.8	32.1	40.1	60.9	69.2	64.1	55.3	41.6	30.9	22.3	(30)
(31)		5%	WB	5.8	4.5	17.0	28.0	35.2	48.9	58.4	55.9	49.6	38.4	28.2	15.8	(31)
(32)			MCDWB	6.3	5.0	17.7	28.7	36.6	54.2	64.0	60.0	52.4	39.3	28.8	16.2	(32)
(33)		10%	WB	1.0	0.1	12.6	24.8	33.5	45.7	55.8	53.9	47.5	36.2	25.7	11.0	(33)
(34)			MCDWB	1.6	0.8	13.1	25.5	34.4	49.9	61.4	57.5	49.6	37.1	26.7	11.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.3	13.8	15.8	16.2	10.2	11.0	13.1	10.8	9.3	8.8	14.1	13.3	(35)
(36)			MCDBR	17.5	18.0	18.6	15.3	11.4	21.0	21.5	17.4	15.7	9.0	13.8	15.7	(36)
(37)			MCWBR	17.3	18.0	18.5	14.9	9.4	13.5	12.0	10.6	10.7	7.9	13.6	15.6	(37)
(38)		5% WB	MCDWB	17.4	18.5	18.8	15.5	11.1	20.7	20.3	15.3	13.8	8.5	14.4	15.7	(38)
(39)			MCWBR	17.3	18.4	18.8	15.3	9.1	13.7	12.5	10.5	10.2	7.6	14.3	15.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.192	0.241	0.266	0.335	0.360	0.345	0.369	0.347	0.319	0.281	0.213	0.155	(40)	
(41)		taud	2.058	2.165	2.187	2.012	2.076	2.380	2.429	2.496	2.526	2.451	2.110	1.943	(41)	
(42)		Ebn at Noon	184	240	273	268	271	280	270	267	256	228	167	145	(42)	
(43)		Edh at Noon	12	22	30	44	45	34	32	28	22	17	11	8	(43)	
(44)	All-Sky Solar Radiation	RadAvg	120	432	1000	1667	1901	1908	1756	1284	763	323	132	60	(44)	
(45)		RadStd	9	18	59	97	165	190	124	90	68	39	18	5	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page